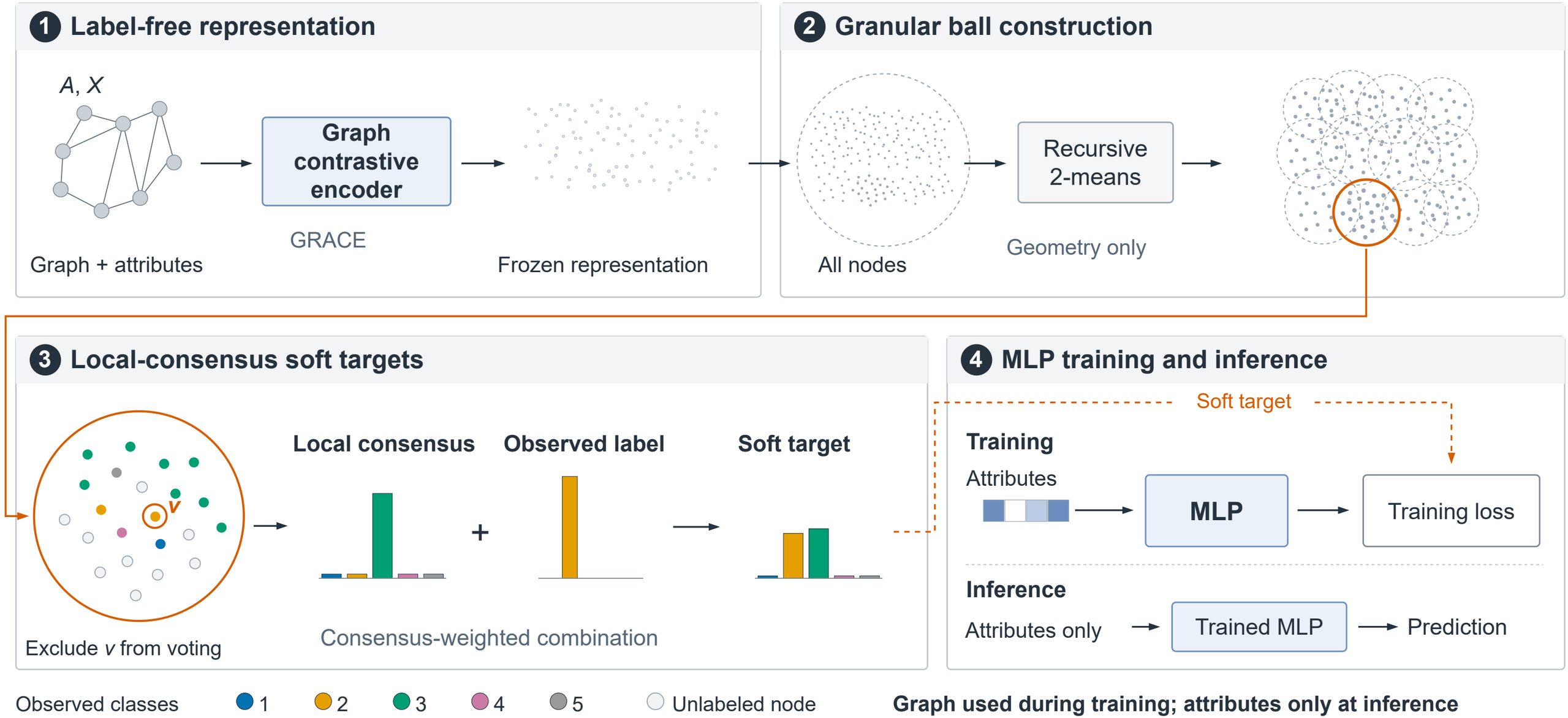

1 Label-free representation
A, X
Graph contrastive encoder
GRACE
Graph + attributes
Frozen representation
2 Granular ball construction
Recursive 2-means
Geometry only
All nodes
3 Local-consensus soft targets
v
Exclude v from voting
Local consensus
Observed label
Soft target
Consensus-weighted combination
4 MLP training and inference
Soft target
Training
Attributes
MLP
Training loss
Inference
Attributes only
Trained MLP
Prediction
Observed classes
1
2
3
4
5
Unlabeled node
Graph used during training; attributes only at inference